

NGUYEN PHAN HOANG PHUC

Data Engineer Intern

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SUMMARY

Final-year student in **Management Information Systems** with a strong **business-first mindset** and a deep curiosity for how digital ecosystems operate. I strongly believe in understanding the business landscape and operational needs before applying technological solutions. Highly capable of **translating complex business requirements into precise data questions** and designing architectures to answer them. While proficient in building scalable data pipelines (**Python, SQL, PySpark**) and modern data modeling (**dbt, Airflow**), my ultimate goal is to leverage these tools to drive actionable insights. Seeking a **Data Engineer Intern** role to bridge the gap between business strategy and robust data infrastructure in a fast-paced environment.

TECHNICAL SKILLS

Programming : Python, SQL, PySpark
Data Engineering : Apache Spark, Apache Airflow, dbt, ETL/ELT Pipelines, Incremental Processing
Databases : Cassandra, MySQL, SQL Server, BigQuery, DuckDB
Cloud : AWS (EC2, S3, IAM), GCP (GCS, BigQuery, Looker)
Tools : Docker, Git, Linux, Tailscale VPN, Tableau

PROJECTS

Customer Behavioral Analytics Pipeline 11/2025 – Present
github.com/hoangf384/Bigdata

- Engineered a containerized **PySpark batch ingestion pipeline** to extract and unify multi-format raw datasets (JSON, CSV, Parquet) from diverse operational modules.
- Architected a scalable **Medallion data workflow (Bronze-Silver-Gold)**, utilizing **Google Cloud Storage** as the **Bronze data lake** by converting raw logs into optimized Parquet files.
- Orchestrated **ELT transformations** in **BigQuery** using **dbt**, modeling **external GCS tables** into staging, intermediate (**Silver**), and business-ready mart tables (**Gold**).
- Integrated **Gold-layer mart tables** with **Looker Studio** or **Metabase**, delivering high-performance, automated Business Intelligence dashboards for downstream analytics.

Job Board Analytics Pipeline 01/2026 – 02/2026
github.com/hoangf384/dataengineering

- Built a data pipeline simulating a **Job Board platform**, answering the question: “*How much did a recruitment campaign spend and how many applications did it generate?*”
- Designed a **Cassandra + MySQL** architecture — Cassandra for high-throughput event ingestion and MySQL for business dimensions and aggregated analytics.
- Developed a **PySpark incremental ETL** using **watermark metadata** to process only new tracking events each batch. Built a **data generator module** producing 10,000 records every 5 minutes to simulate continuous traffic, orchestrated hourly via **Apache Airflow DAGs**.
- Deployed **Dockerized services** across **3 AWS EC2 instances**, connected via **Tailscale VPN Mesh** and **VPC (Security Group)**, with centralized logging on **S3** and **IAM Roles** for credential-free access.

E-commerce ETL Data Pipeline 08/2025 – 09/2025
github.com/hoangf384/crawling-data

- Engineered an automated **Python ETL pipeline** using **Selenium** to extract dynamic **E-commerce** product catalogs, effectively bypassing lazy-loading and complex JavaScript rendering.
- Architected a **Star Schema** data warehouse (Fact/Dimension) to consolidate scattered technical specifications, enforcing strict referential integrity (PK/FK) across 8+ tables.
- Executed rigorous **Data Quality** processing to cleanse, deduplicate, and standardize raw web extractions, ensuring absolute data accuracy for downstream analytics.
- Deployed a robust ingestion architecture featuring comprehensive **Error Logging** for pipeline monitoring, utilizing **Docker Compose** to orchestrate an isolated and stable **MySQL** storage layer.

EDUCATION & QUALIFICATIONS

Management Information Systems | HUB 10/2022 – 03/2026

- Specialization: Business Data Science GPA: 3.2/4.0
- Semi-finalist - Data Analytics Competition 2024 (Organized by HUB University).

Google Data Analytics Professional Certificate

VSTEP – Level 4 (B2 CEFR)